

Do Explanation Signatures Detect Classification Errors? A Controlled Comparison of Graph and Token Explainers

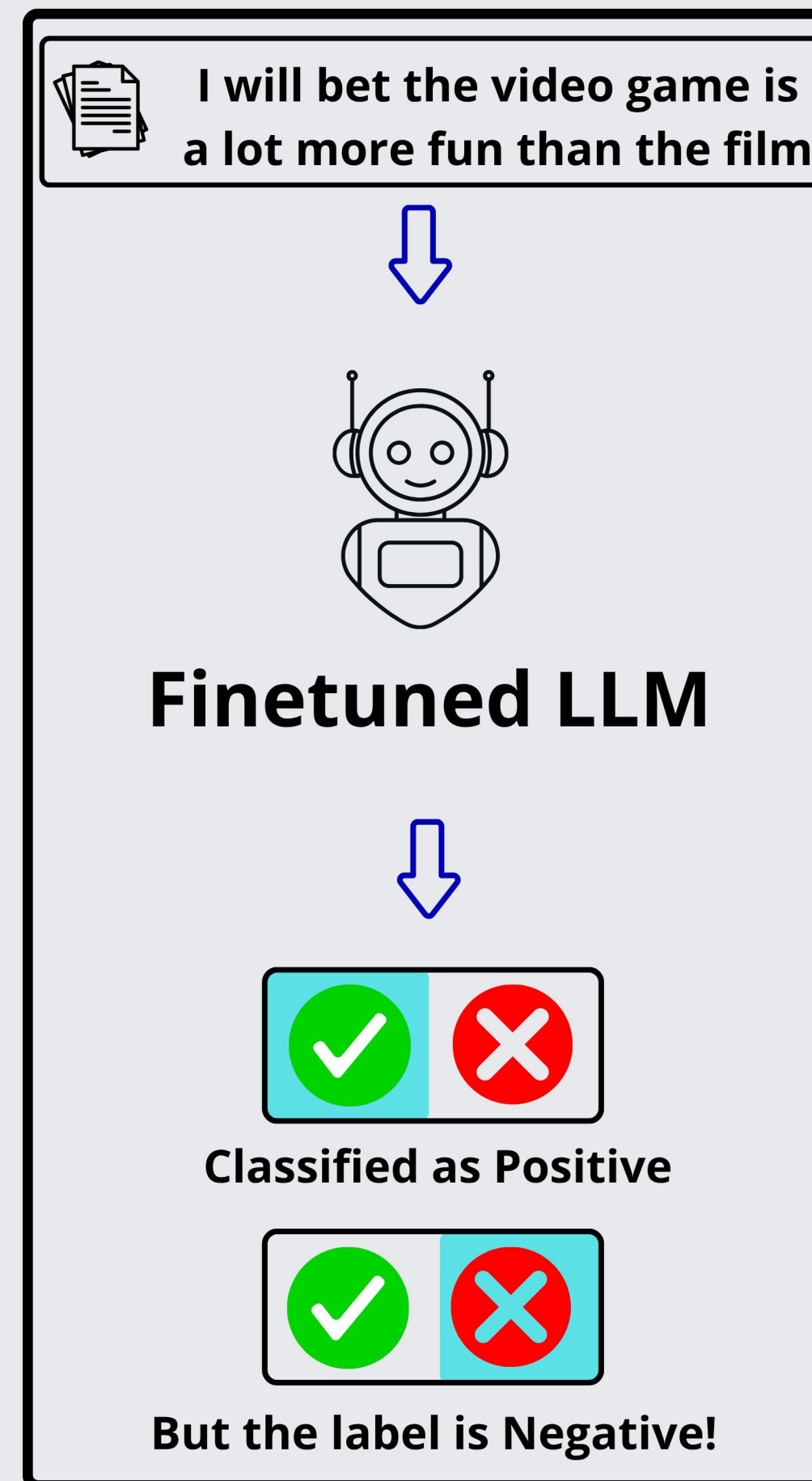
How we compare graph and token explanations on equal terms

Can model explanations reveal when a prediction is wrong?

Explanation features from graphs and from tokens carry information about correctness, but none beyond what the model's own confidence already provides.

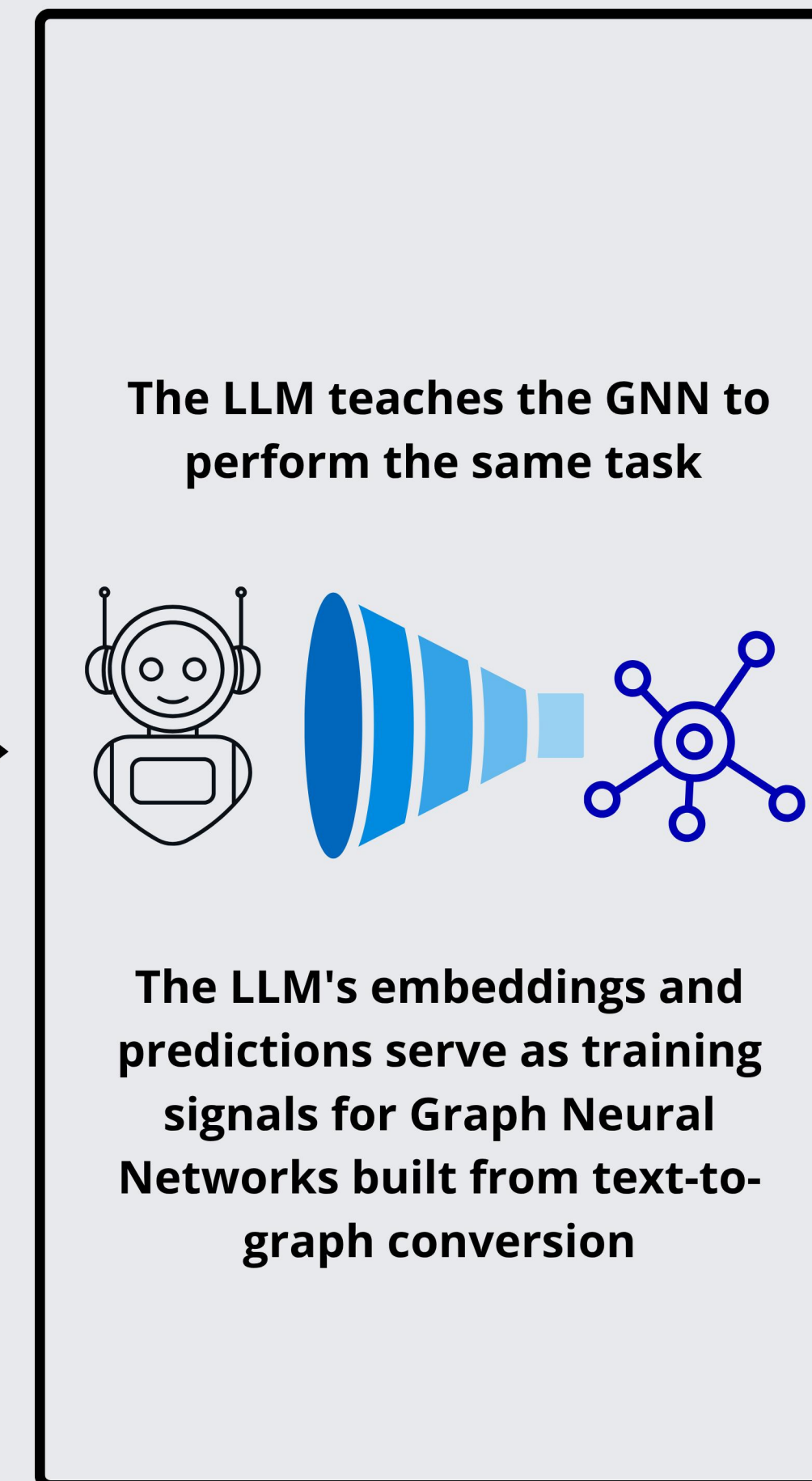
Within the budget every explanation is sufficient and not necessary. A distilled graph surrogate shares its teacher's errors and is confident in them; explanations help it only where that confidence was damaged.

LLM Finetuning



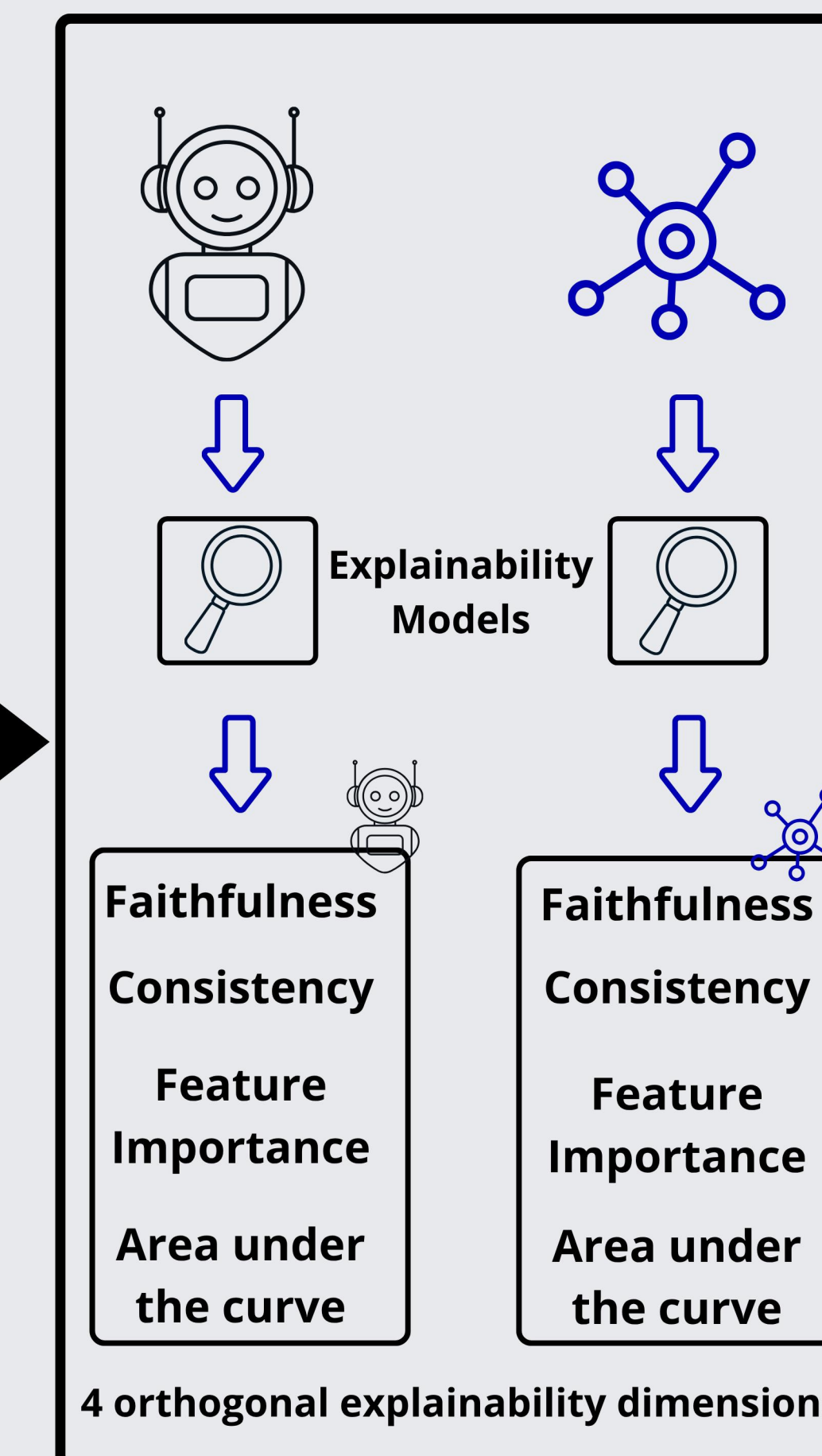
We fine-tune an LLM for text classification and record both its predictions and the ground-truth labels.

LLM as a Teacher



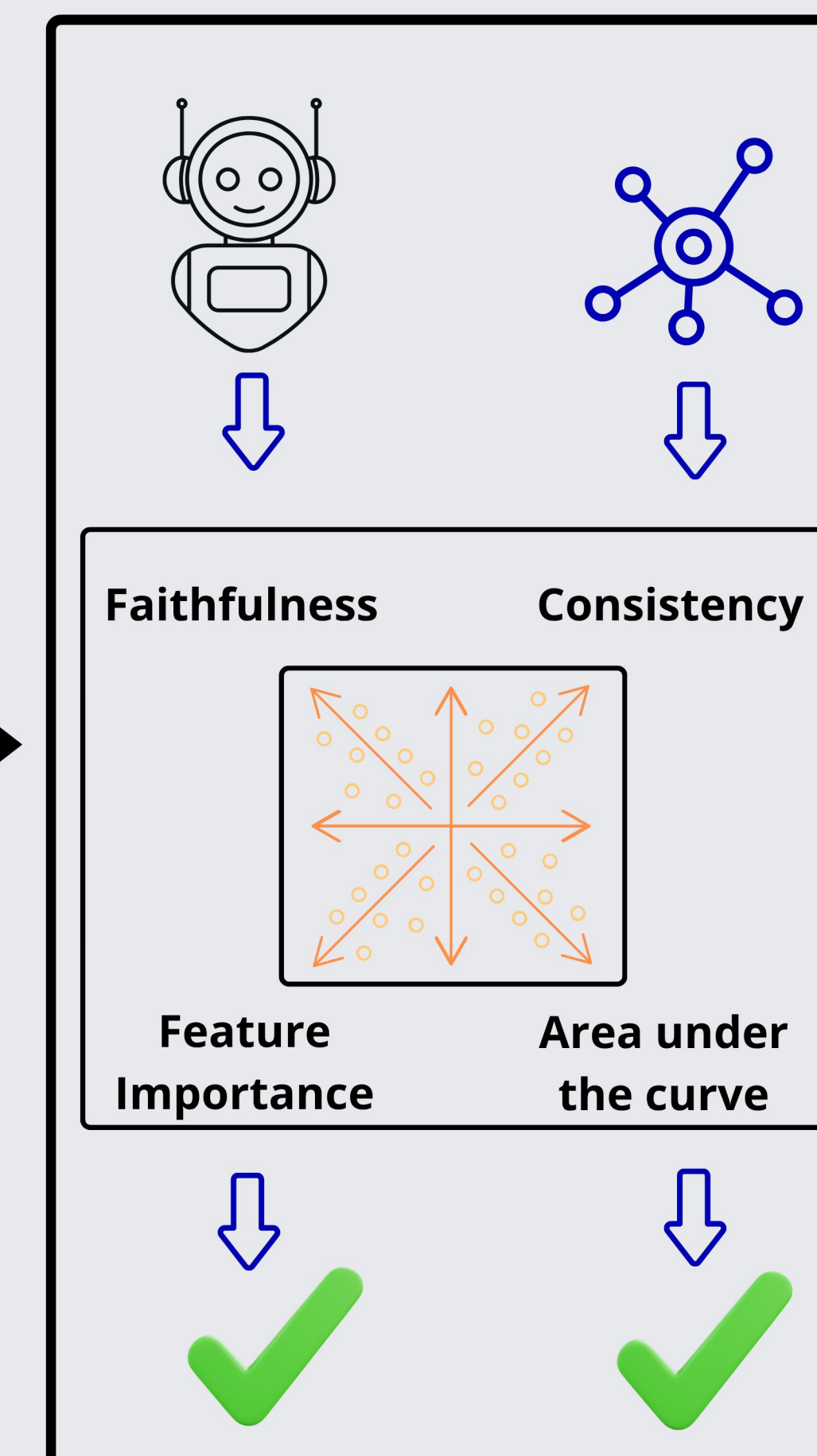
The GNN learns to mimic the LLM for that given task

Get the Explanations



We get patterns from the LLM and GNN explanations across 4 orthogonal dimensions

Logistic Regression



No explanation signature, from graphs or from tokens, detects errors beyond the language model's own confidence